

Compiler Construction Short Answers

A compiler is a software program that converts source code written in a high-level programming language into machine code or executable code that a computer can understand. A compiler translates the whole program at once and generates an executable file, while an interpreter translates and executes the program line by line without creating a separate executable file. A token is the smallest meaningful unit of a program recognized by the compiler, such as keywords, identifiers, operators, and constants. Syntax analysis, also called parsing, is the phase of compiler construction in which the compiler checks whether the source code follows the grammatical rules of the programming language. A symbol table is a data structure used by the compiler to store information about identifiers, including variable names, data types, scope, and memory locations. Code optimization is the process of improving the intermediate code to make the final program execute faster, use less memory, and perform more efficiently.